

brain-music

Live EEG-Driven Music & Visuals from a \$200 Headset

Ashrit Verma · May 14, 2026

AI disclosure: this slide deck (layout, prose, diagrams) was drafted with the assistance of an AI coding agent (Anthropic Claude). The underlying project, code, and engineering decisions are the author's own; AI was used as a writing and design tool for these slides.

The one-liner

A single Python process reads EEG from a **Muse 2** over Bluetooth, generates **continuous music** with **Google Lyria RealTime**, and renders **continuous visuals** in a Three.js shader — all modulated in real time by what your brain is doing.

One prompt drives both the music and the visuals.

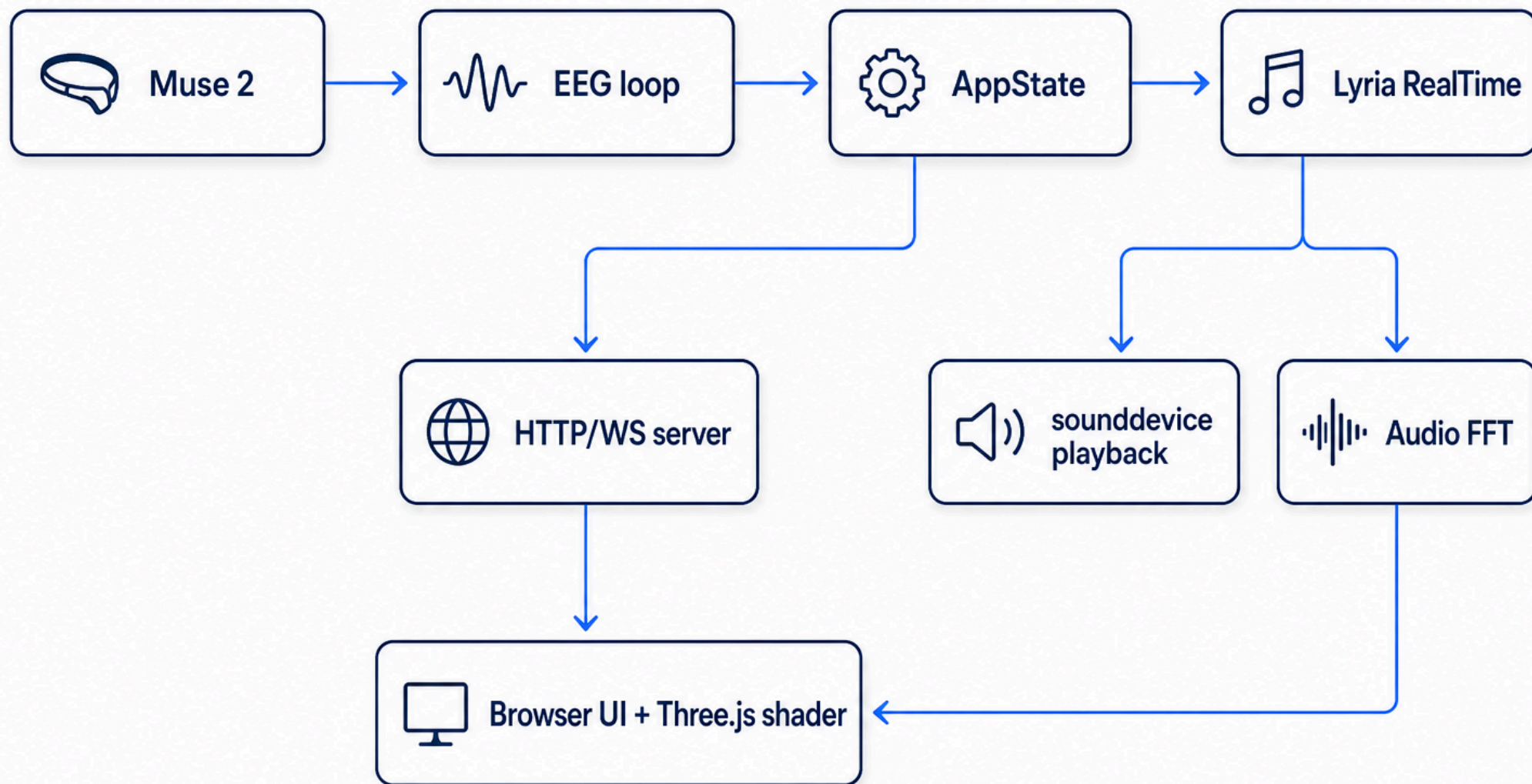
The hardware: Muse 2

Interaxon Muse 2 — a ~\$200 consumer wellness headband.

Key features:

- **4 dry electrodes** — TP9, AF7, AF8, TP10
- **256 Hz** sample rate per channel
- **Bluetooth Low Energy** — no cables, all-day wearable
- **No gel, no skin prep** — put it on and go
- **Onboard accelerometer + gyroscope** (unused here)
- **Lithium battery** — ~5 hr session life





Mapping: EEG → Lyria

EEG feature	Lyria knob	Range	Intuition
α (alpha)	brightness	0 – 1	eyes-closed relaxation → brighter timbre (treble lift)
β (beta)	density	0 – 1	active focus → denser arrangement, more concurrent layers
θ (theta)	temperature	0.5 – 1.9	drowsy / meditative → wilder excursions inside the prompt
asymmetry	bpm	55 – 160	right-frontal lean → faster; left-frontal lean → slower

Asymmetry = frontal alpha asymmetry: $\tanh(\log \alpha_{AF8} - \log \alpha_{AF7})$. Balance of alpha power between the left and right frontal lobes. Idles ≈ 0.5 ; positive values mean less left-frontal alpha (approach motivation).

Global **SENSITIVITY_GAIN = 2.0** — amplifies deviation from the neutral 0.5 midpoint *before* lerping into Lyria's range, so mid-range brain shifts reach the audibly-extreme zone instead of dying in tanh's middle.

The visualizer

Same `AppState` features drive Three.js shader uniforms. Every **15–30 s** the shader cross-fades between four regimes — brain modulates intensity *inside* whichever regime is active.

calm

subtle drift, image stays
mostly stable

tunnel

log-spiral zoom into infinity

kaleidoscope

angular fold into 6–12 mirror
segments

ripple

concentric water-waves from
center

Seed evolver — every ~24 s, a Claude-templated drift descriptor rewrites the prompt and Imagen regenerates the seed texture. The image at minute 5 is downstream of where your brain has been drifting since minute 0.

Latency sources

Stage	Latency	Driven by
Muse 2 BLE + BrainFlow buffering	~30 ms	Packet-level BLE
Welch PSD window	~250–500 ms	<code>WINDOW_SIZE=256</code> ≈ 1 s; median lag for change to register
EMA smoothing	~150–300 ms	<code>SMOOTHING_ALPHA=0.30</code> (~3-tick time constant @ 4 Hz)
Perform tick → Lyria push	0–250 ms	<code>PERFORM_TICK_S=0.25</code>
Lyria chunk boundary wait	0–2000 ms (mean ~1 s)	Dominant term — config applies on the <i>next</i> chunk
Lyria network RTT	~50–150 ms	One WS round-trip per push
Audio output buffer	~20–40 ms	CoreAudio (+ BlackHole if routing to Logic)
End-to-end (brain → audible change)	~1.6 – 2.8 s	Floor when push lands at chunk boundary; ceiling when right after one

Problems encountered

- **Calibration drift across the day.** Same headset, same fit, different time of day → the 8 s baseline μ/σ diverges noticeably. Without re-calibrating, α and β saturate one side and the music pegs. Today: manual `r` / browser **Recalibrate** button. *Want: rolling percentile baseline, auto-managed.*
- **Lyria content filter** silently kills prompts naming an artist/style/album → built a **Claude prompt-guard** that rewrites filtered prompts into pure sonic descriptors (drums, instrumentation, era, vibe).
- **Muse 2 + macOS CoreBluetooth** leaves stale handles after teardown → mandatory 0.3 s + 2 s sleeps in `Board.stop()`, plus `muse2 bt-reset` operator command for nuclear cases.
- **Reconnect supervisors** for both BLE *and* Lyria WS — 3-attempt linear backoff, both can drop at any time.
- **Lyria 2-second chunk boundary** is the floor on responsiveness (see latency slide).

The recalibration problem, in detail

Symptom: start a session in the morning, alpha bar sits at 0.30, music is restrained. Start a second session at 11 PM with the same prompt, alpha sits at 0.85 — music is over-bright and pegging temperature.

Cause: the 8 s baseline only sees that moment's μ/σ . Time-of-day, hydration, recent screen exposure, headset fit micro-shifts all change the noise floor and the log-band-power distribution.

Today: press `r` (or browser **Recalibrate**) — but the user has to *know* it drifted.

Proposed fix: rolling adaptive baseline.

- Reuse the `_AdaptiveBaseline` pattern from `audio/fft.py` (attack 0.30 / release 0.005).
- Track μ over a 60–120 s window continuously.
- Optional UI: show a small drift meter so the operator can see when re-baselining is warranted.

Future improvements

Stage	Today	Proposed
Welch window	1 s @ 256 Hz	Overlap-and-add at 50% hop → ~500 ms lag. Or AR PSD (Burg) on a sliding 0.5 s window → sub-100 ms.
EMA smoothing	$\alpha=0.30$ fixed	Adaptive α : large on direction changes (Kalman-style), small in steady state. ~½ the lag on transitions.
Lyria chunk boundary	0–2 s	Speculative push : tiny AR(2) on the EEG feature, push <i>predicted</i> config one tick early.
Sensitivity	global GAIN=2.0	Per-user auto-tune from the calibration tail's σ .
Mapping curves	single linear lerp	Per-axis curves (logistic for density, gamma for brightness).
Audio features	mono FFT	Stereo L/R centroid difference → panning-aware visuals.

Combined latency floor: **~0.8 s** → **~0.3–0.5 s** — brain → audible change inside one breath.

Future features & how I'd build them

- **Multi-modal fusion** — Whoop HRV, breathing belt, eye-tracker all feed the same `AppState` as new features. *Build:* one adapter module per device; mapping + visualizer untouched.
- **Personalized mapping profiles** — auto-learn the axis a given user controls best and amplify it. *Build:* offline analysis on session logs → per-user `mapping.json` overlay.
- **Calibration-free onboarding** — replace the 8 s capture with auto-fire on SNR + fit stability. *Build:* SNR gate + the rolling baseline from slide 9.
- **Session memory** — save the evolved prompt trail + EEG trace per session; replay later. *Build:* append-only JSON log + a `muse2 replay <session-id>` CLI.
- **Multi-user "duet" mode** — two headsets, one Lyria session, params are an L2-weighted blend. *Build:* `AppStateGroup` aggregator + UI to weight contributors.

Beyond music — channeling the same chain into smart home

The course's purview is music; however, I was inspired!

Building this project directly inspired the next one: **wearable telemetry** → **smart home environment**.

- Same Muse 2 features + **Whoop HRV** + sleep-stage telemetry + ambient sensors → modulate the room itself.
- **Lights** warm + dim when β drops below baseline for >2 min (winding down).
- **HVAC** nudges 1 °F warmer when θ rises late evening.
- **Speakers** cross-fade to ambient when blink rate doubles (eye fatigue).
- Output side is **MQTT + Home Assistant** — much easier to wire than Lyria.

The headband becomes a **context broker for the room**, not just a music controller. Same **AppState** abstraction, different consumers downstream.

What I learned

- **Real-time generative audio + a 2 s chunk model are fundamentally in tension.** Mapping design matters more than push rate.
- The right level for a "brain feature" is **not** raw band power — it's `log → EMA → baseline-normalize → soft-clip`. Operator-controlled floor and ceiling.
- **Consumer EEG is good enough for expressive coupling.** The limit is mapping design + the generative model's response time, not the electrodes.
- A single shared `AppState` + asyncio races beats a multi-process IPC architecture for this scale — easier to reason about, easier to debug, easier to ship to Railway.
- The hardest bug was never the math. It was always **macOS Bluetooth state hygiene.**

Thanks

Questions?

Ashrit Verma · github.com/AshritVerma/brain-music

Why this mapping (and not some other one)

Three explicit design choices, each load-bearing:

- **$\alpha \rightarrow \text{brightness}$, $\beta \rightarrow \text{density}$** . Both move continuously and Lyria responds within ~ 2 s — the right cadence for slow EEG modulation. Discrete knobs (key, scale) wouldn't track band-power drift; continuous ones do.
- **$\theta \rightarrow \text{temperature}$** gives the "*trance state pushes the model out of the safe basin*" feel the audience expects from a brain-driven music box. Theta is the lowest-frequency band; coupling it to model entropy is metaphorically correct *and* sounds right.
- **$\text{asymmetry} \rightarrow \text{bpm}$** (not a band power) because asymmetry is the **only feature centered at 0.5 with symmetric variance**. Band powers after tanh are bottom-heavy $\rightarrow$ bpm would drift to one end. Asymmetry idles bpm at the middle of [55, 160], which is musically neutral.
- **$\text{SENSITIVITY_GAIN} = 2.0$** because tanh of a $z=\pm 2$ deviation squashes to ~ 0.96 . A 1:1 mapping wastes Lyria's envelope; gain=2.0 pulls mid-range brain shifts into the audibly-extreme zone.

Server / API traffic per minute

Steady-state, one user, default config:

Call	Rate	Notes
<code>set_music_generation_config</code> → Lyria	240 /min	4 Hz, one per <code>eeg_tick</code>
Lyria → us (audio chunks)	~30 /min	One ~2 s stereo chunk every ~2 s
WS snapshot → each browser	1,200 /min/client	<code>SERVER_BROADCAST_HZ = 20</code>
Imagen seed evolver	~2.5 /min	1 per 12 chunks, ~150/hr
Claude prompt-polish	~2.5 /min	Paired with each evolver cycle
Claude prompt-guard rewrite	≤ 1 /session	Only if Lyria filters the initial prompt

Cost reality check: ~\$3/hr on Imagen alone at default cadence. Lyria session billed per second of WS connection. Set `--evolve-chunks 0` to skip the evolver entirely.

EEG feature pipeline

Per 1 s window (256 samples @ 256 Hz), 4 Hz output rate:

1. **Welch PSD** per frontal channel (AF7, AF8) — Hanning window, 50% overlap.
2. **Band powers** — θ (4–8 Hz), α (8–12 Hz), β (13–30 Hz).
3. **Log-compress** — band powers are roughly log-distributed; $\log_{10}(x + 1e-6)$.
4. **EMA smooth** — $\alpha=0.30$ (~3-tick / ~750 ms time constant).
5. **8 s baseline calibration** at session start → per-feature μ , σ .
6. **tanh z-score normalize** — $0.5 * (\tanh((x - \mu)/\sigma) + 1)$ → soft-clipped $[0, 1]$.
7. **Frontal alpha asymmetry** — $\tanh(\log \alpha_{AF8} - \log \alpha_{AF7})$ → idle ≈ 0.5 .
8. **Discrete triggers** — blink (peak-to-peak on AF7/AF8 > 1000 μ V), jaw (HP burst > 220 μ V).

Output: four normalized $[0, 1]$ features + two triggers, refreshed at 4 Hz, written to *AppState*.